Deep Learning Lab Minor Project

Title: Deep Learning–Based Classification of Imbalanced Wireless Capsule Endoscopy (WCE) Datasets

Objective

To design, train, and evaluate deep learning models for **gastrointestinal disease classification** using **imbalanced WCE datasets**, while applying **under-sampling, data augmentation-based over-sampling, transfer learning, and intelligent learning rate control**.

Learning Outcomes

After completing this lab, students will be able to:

1. Understand **class imbalance issues** in medical image datasets
2. Apply **under-sampling and augmentation-based over-sampling**
3. Implement **transfer learning with layer freezing and fine-tuning**
4. Use **adaptive / intelligent learning rate control**
5. Evaluate models using **medical classification metrics**

Dataset

- **Kvasir-Capsule Dataset (Imbalanced)**
- **KVASIR v2 Dataset (Balanced, for comparison)**
- **CVC-ClinicDB (Related GI Endoscopy)**
- **ETIS-Larib Polyp DB**
- **KID Dataset (Capsule Endoscopy)**

Lab Tasks

Task 1: Dataset Exploration and Imbalance Analysis

1. Load the Kvasir-Capsule dataset
2. Plot class-wise image distribution
3. Identify majority and minority classes
4. Discuss why class imbalance is critical in medical diagnosis

Deliverable:

- Bar chart of class distribution
- Short explanation (5–6 lines)

Task 2: Under-Sampling (Majority Class Control)

1. Apply **random under-sampling**
2. Reduce majority classes to a fixed threshold (e.g., 200 images/class)

3. Maintain all minority class samples

Goal:
Prevent model bias toward common GI conditions

Deliverable:

- Updated class distribution plot
- Observation on data loss vs balance

Task 3: Data Augmentation–Based Over-Sampling

Apply augmentation **only to minority classes**:

Recommended augmentations:

- Horizontal flip
- Rotation (±20°)
- Width & height shift (0.2)
- Zoom (0.2)
- Any other New Techniques.

Increase minority classes up to the same threshold.

Deliverable:

- Before/after augmentation sample images
- Updated dataset size summary

Task 4: Data Pre-Processing

1. Resize images to **224×224 or 336×336**
2. Normalize pixel values to **[0,1]**
3. Split dataset:
    - Train: 70%
    - Validation: 15%
    - Test: 15%

Task 5: Model Design Using Transfer Learning

Choose **any three**:

- EfficientNet-B0 to B7
- EfficientNey-V2
- MobileNet
- LightCNN29
- ResNet101v2
- InceptionV3

Steps:

1. Load ImageNet-pretrained model

2. Freeze early layers
3. Replace final classification layer
4. Apply dropout + L2 regularization

Deliverable:

- Model summary
- Number of trainable vs frozen parameters
- Comparative study between three models.

Task 6: Intelligent Learning Rate Control

Implement **adaptive learning rate control**, such as:

- Reduce LR on validation-loss plateau
- Cosine decay
- Performance-based LR reduction

Deliverable:

- Learning rate vs epoch plot
- Training vs validation loss curves

Task 7: Model Training and Evaluation

Train the model under **three settings**:

1. Without imbalance handling
2. With under-sampling only
3. With under-sampling + augmentation

Evaluate using:

- Accuracy, Precision, Recall, F1-score,

Deliverable:

- Comparison table
- Confusion matrix
- Short analysis (8–10 lines)

Assessment Rubric

Component	Marks
Dataset analysis & imbalance handling	20
Model design & transfer learning	25
Learning rate control	15
Evaluation & comparison	25
Report clarity & interpretation	15
Total	**100**